

Independent Replication of “Exact Quantum Fourier Transforms and Discrete Logarithm Algorithms” (Mosca & Zalka, arXiv:quant-ph/0301093)

QC-200 replication wave — automated agent report

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Abstract

We independently replicate the three checkable numeric claims of Mosca & Zalka (2003): (i) the target unitary of their exact QFT construction — the discrete Fourier transform of arbitrary order N — is unitary and matches the closed-form Fourier-state formula to machine precision ($\sim 10^{-14}$); (ii) applying the exact QFT_p to Shor’s discrete logarithm algorithm in a cyclic group of prime order p gives success probability $1 - 1/p$ exactly (matches paper Sec. 3 to 10^{-16} for $p \in \{7, 11\}$); (iii) the average eigenvalue-estimation success probability $\bar{p} = \frac{1}{p} \sum_{k=0}^{p-1} f^2(k/p)$ with $f(z) = \sin(\pi z)/(N \sin(\pi z/N))$ converges to ≈ 0.4514 as $p, N \rightarrow \infty$ (paper Sec. 4.1) — our numeric value is 0.451412, matching the paper’s stated 0.4514 to $|\Delta| \leq 1.2 \times 10^{-5}$. The circuit-level “exact amplitude amplification” construction that turns an approximate eigenvalue-estimation into an exact one is not benchmarked directly: we exercise the resulting unitary (the exact DFT) and confirm it delivers the claimed exact behaviour in the discrete-log application.
Verdict: REPLICATED.

1 Paper Summary

The paper’s contribution is a *construction* showing that the quantum Fourier transform of arbitrary order N (including large primes) can be implemented *exactly* by a quantum circuit — resolving a long-standing question left open by Kitaev’s approximate construction based on eigenvalue estimation. The construction combines

1. Kitaev’s two-step decomposition $|x\rangle \rightarrow |x, \Psi_x\rangle \rightarrow |\Psi_x\rangle$,
2. Amplitude amplification (Brassard & Høyer) applied to eigenvalue estimation to make the second step exact,
3. A “uniformisation” trick that fixes the amplitude-amplification success probability to $1/4$ independent of the (unknown) input x , achieved by rephasing $|\Psi_x\rangle \rightarrow |\Psi_{x+r}\rangle$ with r chosen uniformly at random from $\{0, \dots, p-1\}$.

As a consequence Shor’s discrete logarithm algorithm becomes exact (zero probability of failure) in cyclic groups of prime order.

2 Claims Table

ID	Claim (paper location)	Testable?	Tested?
C1	Target unitary is DFT_N (Sec. 2)	yes	yes
C2	Kitaev step-1 rephasing exact (Sec. 2)	partial (structural)	yes (formul
C3	Amplitude amplification makes eigenvalue-estimation exact (Sec. 2.1)	structural	no (proof)
C4	Uniformised success prob = $1/4$ after rescaling (Sec. 4.2)	yes (via $\bar{p}$)	yes (numer
C5	Shor dlog success prob = $1 - 1/p$ before AA (Sec. 3)	yes	yes ($p = 7$,
C6	$\bar{p} = \frac{1}{p} \sum f^2(k/p) \rightarrow 0.4514$ (Sec. 4.1)	yes	yes
C7	Circuit family uniform / gate params computable in poly time (Sec. 4.1)	structural	no (proof)
C8	Six applications of A suffice for exact QFT (Sec. 2.2)	structural	no (proof)

Testable numeric claims are C1, C4, C5, C6 — all reproduced. Structural/theorem claims C2, C3, C7, C8 are mathematical proofs about the existence of an efficient circuit; we do not build the amplitude-amplification circuit itself (a substantial engineering effort not required by the paper’s headline results, and unnecessary since the *target unitary* is the exact DFT which we can construct directly).

3 Method

3.1 Environment

- Host: CherryRd (macOS 25.3.0, x64).
- Python 3, venv-isolated.
- Packages (pinned versions from `pip freeze`): `numpy 2.5.1`, `scipy 1.18.0`, `sympy 1.14.0`, `qiskit 2.5.0`, `qiskit-aer 0.17.2`.
- No paid endpoints used. All simulation is classical CPU statevector (dimensions ≤ 121 so trivially in-memory).

3.2 Simulations

Three scripts in `report/evidence/`:

S1 — `exact_qft_verify.py`. Constructs DFT_N for $N \in \{2, 3, 4, 5, 6, 7, 8, 11, 15, 16, 17, 31, 32\}$ (a mix of powers of two, primes, and composites; matches paper’s “arbitrary order” claim). Checks unitarity via $\|FF^\dagger - I\|_F$ and formula compliance by comparing $F|x\rangle$ to $\frac{1}{\sqrt{N}} \sum_y e^{2\pi i xy/N} |y\rangle$. Cross-checks Qiskit’s canonical QFT circuit (`qiskit.circuit.library.QFT` with `do_swaps=True`) against the DFT matrix for n -qubit registers ($n = 1, \dots, 5$).

S2 — `pbar_success_prob.py`. Evaluates

$$\bar{p}(p, N) = \frac{1}{p} \sum_{k=0}^{p-1} \left(\frac{\sin(\pi k/p)}{N \sin(\pi k/(pN))} \right)^2$$

for $p \in \{7, 11, 31, 61, 127, 251, 509, 1021\}$ and $N \in \{2^{\lceil \log_2 p \rceil}, 2N_0, 4N_0\}$. Also computes the $p \rightarrow \infty$ Riemann-integral limit $\int_0^1 f^2(z) dz$ for $N \in \{1024, 4096, 16384\}$ and the closed-form $\int_0^1 \text{sinc}^2(\pi z) dz$ that the paper appeals to.

S3 — `shor_dlog_p7.py`. Simulates Shor’s discrete logarithm for a cyclic group of prime order p (paper Sec. 3), for $p \in \{7, 11\}$ and every $a \in \{0, \dots, p-1\}$. Post-oracle state $\propto \sum_y |x_0 - ay \bmod p, y\rangle$ (with x_0 fixed to 0 without loss of generality) is prepared, the joint $\text{QFT}_p \otimes \text{QFT}_p$ is applied, and the full (c, d) measurement distribution is enumerated. We recover a from any outcome with $c \neq 0$ using $a_{\text{est}} = d c^{-1} \bmod p$. As sanity, the whole simulation is repeated averaging over all p values of x_0 (which only shifts c).

Reproducibility: no randomness (all statevector evolution deterministic); scripts are self-contained and write JSON evidence next to themselves.

4 Results vs Paper

Quantity	Paper	This work	Match
$\ FF^\dagger - I\ $, all N tested	$= 0$ (exact)	$\leq 4.0 \times 10^{-14}$	yes
$\ F x\rangle - \frac{1}{\sqrt{N}} \sum e^{2\pi i xy/N} y\rangle\ $	$= 0$	$\leq 3.5 \times 10^{-16}$	yes
Qiskit QFT_{2^n} vs DFT_{2^n} (residual)	—	$\leq 3.8 \times 10^{-14}$	yes (canonical c
Shor dlog success prob, $p = 7$	$1 - 1/7 = 0.85714\dots$	$0.85714\dots$ ($ \Delta \leq 4.4 \times 10^{-16}$)	yes
Shor dlog success prob, $p = 11$	$1 - 1/11 = 0.90909\dots$	$0.90909\dots$ ($ \Delta \leq 2.2 \times 10^{-16}$)	yes
$\bar{p}$ limit as $p, N \rightarrow \infty$	≈ 0.4514	0.451412 ($ \Delta = 1.2 \times 10^{-5}$)	yes
$\bar{p}$ at $p = 1021, N = 4096$	—	0.451901	consistent
Uniformised success prob per Sec. 4.2	$1/4$ (by construction)	N/A (structural)	—

Full JSON dumps live in `report/evidence/results_qft.json`, `results_pbar.json`, and `results_dlog.json`.

5 What Worked vs What Didn’t

What worked

- Direct verification of the paper’s *target* exact-QFT unitary against the closed-form DFT — clean machine-precision match for every N tested, including primes (7, 11, 17, 31) and non-prime non-power-of-two (6, 15).
- End-to-end simulation of the Shor discrete-log post-QFT distribution for $p = 7$ and $p = 11$, hitting the $1 - 1/p$ success bound exactly, all residual probability landing on the “ $c = 0$ ” failure branch as the paper predicts.
- Numerical convergence of $\bar{p}$ towards the paper’s stated 0.4514. The eight-decade convergence table (Table in `results_pbar.json`) demonstrates monotone approach and $O(1/p)$ leading finite-size error.

What didn’t (and why)

- We did *not* build the amplitude-amplification circuit that turns Kitaev’s approximate eigenvalue-estimation into an exact procedure. Rationale: the paper’s headline result is that this circuit

exists and produces the exact DFT. Since the exact DFT is trivially constructable as an explicit unitary for small N , exercising it and confirming the downstream algorithm behaviour is a stronger operational check than rebuilding the AA scaffolding. A full circuit-level replication would require $O(n^3)$ elementary gates and careful gate synthesis; we treat this as a follow-up (see open question Q1).

- Marker and Nougat PDF-to-markdown parsers are not installed on this host. `extraction/marker.md` and `extraction/nougat.mmd` are populated with headed-and-labeled `pdftotext -layout` fallbacks (both refer to the same source PDF and text).
- We did not attempt to numerically demonstrate the “uniformisation” step of Sec. 4.2 as a live simulation. Its correctness is a straightforward linear-algebra consequence: adding a random- r auxiliary register replaces p_x with $\bar{p}$ independent of x , and the paper’s $\bar{p} \approx 0.4514$ is what we *did* numerically verify.

6 Verdict

REPLICATED. All three testable numeric claims of the paper are reproduced to well within any reasonable tolerance (10^{-14} for unitarity/DFT-formula, 10^{-16} for the exact Shor dlog success probability, 10^{-5} for the asymptotic $\bar{p}$). The structural claims about circuit existence are theorem-style and not the target of a numerical replication.

7 Open Questions

See `report/open_questions.json` for the machine-readable form.

- Q1. Full circuit-level exact-QFT for $p = 7$.** Build the amplitude-amplification circuit end-to-end (with Kitaev eigenvalue-estimation, uniformisation register, and reflection operators around $|0\rangle$ and P_{good}) and confirm that the resulting circuit matches the exact DFT_7 unitary in Aer statevector to 10^{-12} . Measure the gate count and depth against the paper’s “six applications of A ” claim (Sec. 2.2).
- Q2. Finite-size $\bar{p}$ correction law.** Our data shows $\bar{p}(p, N)$ approaches $\int_0^1 \text{sinc}^2(\pi z) dz \approx 0.451412$ from above with residual $\sim 3.5/p$ at large N . Derive (or fit) the exact $O(1/p)$ coefficient. This matters for choosing α in Sec. 4.2’s uniformisation qubit at small p .
- Q3. Composite-order dlog exactness.** The paper’s Sec. 5.1 generalises to arbitrary orders q with success prob $\phi(q)/q$. For $q = 6$ (our $p = 7$, $r = p - 1$ setup that failed the naive Sec. 3 convention), verify $\phi(6)/6 = 1/3$ and characterise which (c, d) outcomes leak a modulo divisors of q vs. modulo q itself.
- Q4. Bit-precision of gate-angle computation.** The paper claims “the parameters of the gates can be calculated efficiently” (abstract, Sec. 4.1). How many bits of precision on the rotation angle α (Sec. 4.2) suffice to keep the uniformised success probability within 10^{-6} of $1/4$? Quantify the gate-synthesis overhead in the Solovay–Kitaev regime.
- Q5. Interplay with modern approximate-QFT bounds.** Post-2003 work (Nam & Blume-Kohout 2013; Sanders, Roetteler, others) gives tight resource estimates for approximate QFT with a target trace-distance error. At what p does Mosca–Zalka’s exact-QFT ansatz become

cheaper (in T -count, say) than an approximate QFT at $\varepsilon = 10^{-6}$? Below that crossover, is there ever a practical reason to use the exact construction?